

System Role: You are a highly critical, world-renowned AI Scientist and Peer Reviewer specializing in the evaluation of Large Language Models (LLMs) for ****Precision Question Answering in Agriculture Systems****. You have been given a full research paper to review.

Task: Read the paper provided below. Based on the paper's stated methodology propose a rigorous experimental design setup.

Input Text (Research Paper):

Abstract: To enhance the accessibility and accuracy of plant protection knowledge for agricultural practitioners, this study develops an intelligent question-answering (QA) system based on a large language model (LLM). A local knowledge base was constructed by vectorizing 7000 research papers and books in the field of plant protection, from which 568 representative papers were selected to generate QA data using an LLM. After data cleaning and filtering, a fine-tuning dataset comprising 9000 question-answer pairs was curated. To optimize the model's performance, low-rank adaptation (LoRA) was applied to the InterLM-20B model, resulting in the InterLM-20B-LoRA, which was integrated with Langchain-ChatChat and the local knowledge base to develop the QA system. Additionally, retrieval-augmented generation (RAG) technology was implemented to enhance response accuracy by enabling the model to retrieve relevant field-specific knowledge before generating answers, effectively mitigating the risk of hallucinated information. The experimental results demonstrate that the proposed system achieves an answer accuracy of 97%, highlighting its potential as an advanced solution for intelligent agricultural QA services.

1. Introduction

Plant protection is a critical component of modern agricultural production, encompassing pest and disease management, soil health preservation, and crop yield optimization [1,2]. The rapid advancement of agricultural practices has introduced complex challenges, including the emergence of novel plant pathogens, increasing pest resistance to chemical control measures, and the impacts of climate change on crop productivity [3,4]. Accurate, timely plant protection knowledge is essential. This is particularly true for farmers, who are the primary field practitioners and often lack specialized expertise. We aim to provide them with practical, real-time recommendations. These guidelines are designed to address their daily challenges effectively. However, conventional knowledge acquisition methods, such as expert consultations, printed reference materials, and fragmented online resources, are often time-consuming, require specialized expertise, and may not provide real-time, context-specific recommendations. Moreover, variations in crop species, regional environmental conditions, and pest dynamics necessitate tailored solutions, making generic knowledge dissemination approaches insufficient to address the diverse and evolving needs of the agricultural sector.

In recent years, large language models (LLMs) have demonstrated exceptional natural language understanding and generation capabilities, finding applications in diverse fields such as healthcare, finance, and legal services [5-7]. Their ability to process complex queries and generate contextually relevant responses makes them promising candidates for question-answering (QA) systems. However, applying LLMs to highly specialized domains such as plant protection presents significant challenges. One major limitation is their lack of domain-specific knowledge, as pre-trained models rely primarily on publicly available datasets that often lack high-quality, expert-reviewed agricultural content. Consequently, these models may generate incomplete, outdated, or misleading responses when addressing plant protection issues. Additionally, LLMs are prone to the well-documented hallucination problem, where they produce factually incorrect or unverifiable information.

In plant protection, where accurate and scientifically validated knowledge is essential for effective decision-making, such hallucinations can result in incorrect diagnoses, improper pesticide applications, and ineffective pest management strategies, potentially leading to economic losses and environmental harm [8]. Furthermore, traditional LLMs struggle to adapt to region-specific agricultural conditions, as factors such as climatic variations, soil characteristics, and pest dynamics necessitate dynamic and context-aware responses that generic models fail to provide.

To mitigate these challenges, fine-tuning has been widely explored to enhance LLM performance in specialized domains. However, conventional fine-tuning modifies a large number of parameters, requiring extensive computational resources and substantial storage, making it impractical for organizations with limited hardware capabilities. A more efficient alternative is low-rank adaptation (LoRA), a parameter-efficient fine-tuning method that significantly reduces computational overhead while preserving model performance. LoRA achieves this by introducing trainable low-rank matrices within specific layers of the pretrained model instead of modifying the full parameter set, enabling efficient adaptation to new tasks [9,10]. This approach not only enhances accessibility for domain-specific applications but also ensures that the model retains its core linguistic capabilities. In this study, we leverage LoRA to fine-tune an LLM for plant protection, equipping it with specialized knowledge while maintaining computational efficiency. In addition to fine-tuning, retrieval-augmented generation (RAG) has emerged as a powerful technique for enhancing the accuracy and reliability of LLM responses. RAG integrates information retrieval with text generation, allowing the model to access an external knowledge base when formulating responses dynamically. Unlike traditional LLMs that rely solely on static training data, RAG-based models can retrieve relevant

information from a curated document corpus, thereby reducing hallucinations and ensuring that responses are grounded in authoritative sources [11]. In this study, we construct a local knowledge base consisting of professional scientific literature on plant protection, which serves as the retrieval component of our RAG QA system. This enables the model to reference validated agricultural research and expert-authored content when answering user queries, improving response quality and trustworthiness. The main contributions of this article mainly include the following points:

- We develop an intelligent QA system tailored for the plant protection domain, addressing the critical need for precise and accessible agricultural knowledge. By leveraging natural language processing (NLP), we bridge the gap between complex scientific research and practical decision-making, enabling farmers and agricultural practitioners to obtain expert-level insights efficiently. The system is designed to accommodate diverse user needs, providing customized recommendations based on varying geographical conditions and crop-specific requirements.
- We integrate advanced fine-tuning and retrieval strategies to optimize LLM performance for plant protection applications. By combining LoRA-based fine-tuning with RAG-enhanced retrieval, we mitigate common limitations of pre-trained models, such as hallucinations and domain knowledge deficiencies. Additionally, we construct a local domain-specific knowledge base composed of peer-reviewed scientific publications, ensuring that the model's responses are accurate, authoritative, and explainable. This hybrid approach significantly enhances the reliability of LLM-generated answers in an agricultural context.
- We contribute a high-quality plant protection dataset containing over 9000 question-answer pairs, covering fundamental plant protection principles and expert-verified knowledge extracted from academic papers. This dataset serves as a valuable resource for training and evaluating domain-adapted language models. By fine-tuning our model on this dataset using LoRA, we achieve substantial improvements in domain-specific comprehension and response accuracy, demonstrating the effectiveness of our approach in equipping LLMs with specialized agricultural knowledge.

3.2. Methods

3.2.1. InternLM Model

InternLM [34] was developed through a collaborative effort by the Shanghai Artificial Intelligence Laboratory, SenseTime, the Chinese University of Hong Kong, Fudan University, and Shanghai Jiao Tong University. Built upon the Transformer architecture, the model is designed to tackle complex multi-task natural language processing (NLP) challenges. Pre-trained on diverse high-quality datasets, InternLM further enhances its language understanding and generation capabilities through self-supervised learning. The model is optimized for multilingual scenarios. It not only processes Chinese fluently but

also exhibits strong cross-lingual capabilities, particularly in applications such as dialog generation and question-answering systems. Leveraging its large-scale parameters, InternLM exhibits strong adaptability and efficiency in complex reasoning tasks. In terms of architecture optimization, InternLM integrates a variety of innovative technologies to improve the training and reasoning efficiency of the model. First, during training, the model employs a hybrid parallelization strategy that integrates data parallelism, model parallelism, and pipeline parallelism, significantly enhancing distributed training efficiency. In addition, by incorporating a sparse attention mechanism, InternLM effectively reduces computational overhead in processing long text sequences, thereby enhancing inference speed. Notably, the model supports multimodal data fusion, enabling comprehensive analysis of text, image, and audio information while enhancing performance in cross-modal tasks. Finally, self-supervised learning enables the model to train on vast amounts of unlabeled data, further enhancing its generalization ability, particularly in multilingual tasks.

3.2.2. Low-Rank Adaptation

Low-rank adaptation (LoRA) [35] is an advanced fine-tuning strategy specifically designed to efficiently optimize the parameters of large pre-trained language models.

Instead of modifying all parameters, LoRA optimizes model weights by introducing lowrank matrices, which dramatically reduces the number of trainable parameters during finetuning.

Compared to traditional full-parameter fine-tuning methods, LoRA not only lowers computational costs but also preserves the model's core parameters, thereby mitigating the risk of overfitting. Its ability to quickly adapt to new tasks while maintaining overall model stability makes LoRA particularly well-suited for domain-specific applications that require frequent adaptation.

In traditional full-parameter fine-tuning, all model parameters must be updated for a new task. In contrast, LoRA significantly reduces the number of trainable parameters by decomposing the weight matrix into two low-rank matrices. LoRA introduces two low-rank matrices, A and B, with smaller dimensions to approximate the original weight matrix. During fine-tuning, only these low-rank matrices are updated, leaving the original weight matrix unchanged.

Assuming that the weight matrix of a large language model is W_0 , LoRA updates it in the following way:

$$h = W_0x + Wx = W_0x + BAx \quad (1)$$

where $W = BA$. Due to the low rank of A and B, the number of updated parameters is

drastically reduced. In this way, the computational complexity and storage requirements of the model are also reduced.

In this study, we applied LoRA to enhance the performance of large language models

in plant protection question-answering tasks. Compared to traditional full-

parameter finetuning, LoRA updates only the low-rank matrices of specific layers, reducing computational overhead while preserving the model's original knowledge structure. By applying LoRA to the InternLM-20B model, we improved its ability to model professional terminology and domain-specific knowledge in plant protection.

3.2.3. Retrieval-Augmented Generation

Retrieval-augmented generation [36,37] (RAG) leverages the strengths of both retrieval and generation models to enhance the performance of large language models in knowledge-intensive tasks. While traditional generative models can generate fluent text, they often produce misleading or incomplete responses when handling queries requiring specific domain knowledge due to a lack of accurate contextual information. In contrast, RAG incorporates retrieval mechanisms to effectively leverage local knowledge bases during answer generation, thereby producing more accurate and well-supported responses. The RAG framework comprises two key components: the retrieval component and the generation component.

Retrieval Component: Upon receiving a user query, the system employs a predefined retrieval model to search the local knowledge base for the most relevant documents. The retrieved documents are transformed into vector representations using embedding techniques, enabling efficient similarity searches.

Generation Component: Once relevant documents are retrieved, the generation component processes them. It typically consists of a large pre-trained language model that integrates the retrieved documents with the user query to generate responses. This mechanism enables the model to generate responses based on the latest and most relevant information from the knowledge base, thereby enhancing accuracy and comprehensiveness. In the RAG framework, the generation component does not solely rely on the model's internal knowledge but dynamically integrates information from the local knowledge base to generate responses. This approach significantly reduces the 'hallucination' phenomenon commonly observed in traditional generative models when addressing domain-specific queries, reducing the risk of generating inaccurate or misleading information. This study develops a plant protection knowledge question-answering system leveraging RAG technology to enhance answer accuracy and reliability. The system integrates the generative capabilities of a fine-tuned large language model with dynamic retrieval of professional literature, research papers, and technical guidelines in plant protection, ensuring responses that are timely, relevant, and domain-specific. The FAISS vector library is employed to convert knowledge base documents into vector representations. When a user submits a query, it is transformed into a vector and matched against the knowledge base using similarity search, enabling rapid retrieval of the most relevant information. The retrieved documents serve as contextual information,

which, along with the user query, is fed into the generation module to produce responses that incorporate expert plant protection knowledge and accurately address complex plant pest and disease issues. The integration of RAG technology significantly enhances the system's capability to handle multi-domain knowledge and complex agricultural scenarios, delivering more precise and contextually relevant responses to users.

3.2.4. Evaluation Metrics

The evaluation of large language models covers multiple dimensions [38], including language generation ability, task performance, comprehension depth, and logical reasoning. Automatic evaluation employs computational metrics to quantitatively assess both the quality of generated text and overall task performance. Common evaluation metrics include BLEU [39], ROUGE [40], BERTScore [41], QAGS [42], FactCC [43], and FEQA [44], which measure aspects such as text similarity, semantic consistency, and fluency. However, BLEU rely on precise n-gram matching and are less sensitive to semantic equivalence and syntactic variations. Furthermore, although QAGS and FEQA can indirectly measure text fidelity through question-answer generation, their question-generation process often introduces significant uncertainty, and their correlation with human judgment remains limited in practical applications. For these reasons, we have chosen not to use BLEU, QAGS, and FEQA, and instead focus on metrics that more accurately capture semantic and factual consistency. In this study, we employ ROUGE, BERTScore, and FactCC as automatic evaluation metrics to assess model performance. These three methods are both efficient and objective, providing a comprehensive measure of the model's performance on generation tasks. ROUGE primarily measures the overlap between generated and reference text, making it particularly suitable for assessing summarization accuracy and other text generation tasks. BERTScore leverages a deep language model to embed both the generated text and the reference text, then calculates the cosine similarity between these embeddings to capture semantic information. FactCC evaluates factual consistency by using a supervised binary classifier to determine whether the factual statements in the generated text align with authoritative data, effectively identifying hallucinations and factual errors. In summary, these indicators effectively capture text quality and semantic accuracy while reducing the subjectivity and time costs associated with manual evaluation, thereby providing a more reliable basis for the performance evaluation of large-scale language models. ROUGE (Recall-Oriented Understudy for Gisting Evaluation) is primarily used to measure the similarity between generated and reference texts based on N-gram overlap. An N-gram is a contiguous sequence of N items—such as characters, syllables, or

words—in a text, where N is a positive integer denoting the sequence length. ROUGE is widely applied in automatic summarization and machine translation tasks. Common ROUGE variants include ROUGE-N and ROUGE-L. ROUGE-N quantifies the N -gram overlap between generated and reference texts. The equation is:

$$\text{ROUGE - N} = \frac{\sum_{S \in \{\text{Reference Summaries}\}} \sum_{\text{gramN}} \text{Countmatch}(\text{gramN})}{\sum_{S \in \{\text{Reference Summaries}\}} \sum_{\text{gramN}} \text{Count}(\text{gramN})} \quad (2)$$

The numerator of the equation calculates the minimum number of matches of all N -grams between the generated text and the reference text, while the denominator is the total number of N -grams in the reference text.

ROUGE-L is based on the longest common subsequence (LCS) to capture the structural similarity between the generated text and the reference text. The equation is expressed as follows:

$$\text{ROUGE - L} = \frac{\text{LCS}(\text{gen}, \text{ref})}{\text{length}(\text{ref})} \quad (3)$$

where $\text{LCS}(\text{gen}, \text{ref})$ is the length of the longest common subsequence (LCS) between the generated text and the reference text, and $\text{Length}(\text{ref})$ is the length of the reference text; ROUGE-L is mainly used to calculate the similarity between the generated text and the reference text in terms of sentence structure, which is applicable to the scenario of structured information.

BERTScore is a metric designed to measure text similarity. It evaluates the similarity between two sentences by calculating the cosine similarity between the embedding vectors produced by a pre-trained BERT model.

The core idea behind BERTScore is to input two sentences individually into the BERT model, obtain their vector representations, and then compare these vectors. Unlike traditional similarity methods based on static word vectors, BERTScore more accurately assesses sentence similarity by incorporating contextual information and semantic features.

In particular, for sentences that convey similar meanings despite differing in expression, BERTScore more effectively captures their underlying similarity.

$$\text{RBERT} = \frac{1}{|X|} \sum_{x_i \in X} \max_j \text{SjES}_{x_i} \quad (4)$$

$$\text{PBERT} =$$

$$\frac{1}{|S|} \sum_{S_j \in S} \max_i X_i^T X_j$$

$$\frac{1}{|S|} \sum_{S_j \in S} \frac{S_j^T X_j}{\|S_j\|_2 \|X_j\|_2} \quad (6)$$

where X denotes the reference text, X_i denotes the i -th reference text, S denotes the generated text, S_j denotes the j -th generated text, and X^T denotes the transpose matrix.

FactCC is a metric designed to measure factual consistency by determining whether

the generated text is consistent with the reference text. It relies solely on the reference text to verify factual alignment. The core idea behind FactCC is to input each generated sentence

along with the reference text into a supervised classifier that has been trained on labeled

data, indicating whether a sentence is factually consistent. By averaging the classifier's outputs of overall generated sentences, FactCC produces an overall factual consistency

score. The calculation is as follows:

$$\text{CFactCC}(S, R) =$$

$$\frac{1}{|S|} \sum_{s \in S} P(C = 1 | s, R) \quad (7)$$

where $\text{CFactCC}(S, R)$ denotes the factual consistency score for the generated text S given

the reference text R , and $P(C = 1 | s, R)$ represents the classifier's probability that the

generated sentence s is factually consistent with R .

****Required Output: Critical Experimental Design Setup****

Please structure your response into the following three parts:

1. ****Summary of the Paper's Core Method & Evaluation:**** Briefly state the system and the primary evaluation metrics used.

2. ****Identified Flaw/Missed Opportunity in Experimental Design:**** State ****one specific area**** where the paper's experiment could be significantly improved. This must relate to baselines, datasets, or evaluation metrics.

* ****Example areas:**** Missing a necessary domain-specific baseline, using generic metrics when more precise factual consistency checks are available, or failing to test generalization across different agricultural regions.

3. ****Proposed Alternative/Addition (The "AI Scientist's Better Way"):**** Detail the specific experimental element that should have been used instead, or in addition.

* ****If a Baseline is Missing:**** Which specific model (name and type) should have been included?

* ****If a Metric is Weak:**** Which specific, state-of-the-art metric (e.g., RAGAS, LLM-as-a-Judge, FEQA) should have been used, and why is it superior in this agricultural domain?

* ****If a Significance Test is Missing:**** Which statistical test (e.g., T-test, ANOVA) should have been run to prove the results were statistically robust?

6. Conclusions

This paper presents the development of an intelligent question-answering system tailored for the plant protection domain, leveraging large language model fine-tuning and retrieval-augmented generation (RAG). Extensive experiments demonstrate the efficacy of this approach in enhancing answer accuracy, professionalism, and interpretability. By employing low-rank adaptation (LoRA), the pre-trained model is efficiently infused with specialized domain knowledge without a significant increase in computational overhead. Simultaneously, the integration of RAG, which facilitates real-time retrieval of professional literature and agricultural data, markedly reduces the occurrence of "hallucinated" information, thereby bolstering the credibility of the responses. Comparative analyses reveal that post fine-tuning and real-time retrieval optimization, the system exhibits robust performance and delivers high-quality answers to challenges such as agricultural pest control, crop growth regulation, and regional agricultural promotion. Nonetheless, certain limitations persist within this research. The constructed plant protection dataset, while encompassing fundamental theories and specialized knowledge, requires further expansion in both breadth and depth, particularly concerning niche areas. The system's heavy reliance on external literature repositories means that the quality and update frequency of these sources directly influences response efficacy. Moreover, the current focus on textual data processing, without the integration of multimodal information like images and videos, may constrain the system's applicability in future agricultural contexts. Despite these challenges, this study offers a viable technical pathway for domain-specific question-answering systems, underscoring the significant role of combining large language model fine-tuning with real-time information retrieval in reducing "hallucination" issues and enhancing domain adaptability. Future endeavors should aim to expand the dataset's scope, enhance the authority of external knowledge bases, and explore the fusion of multimodal data. Such efforts will serve to refine system functionality and elevate overall performance. In essence, this work introduces innovative perspectives and practical insights for constructing intelligent question-answering systems in the plant protection sector, contributing positively to the advancement of agricultural informatization and informed decision-making.